Year 9 Biology Homework #1
Name:_____

Due_____

HUMAN NERVOUS SYSTEM

➢Nervous system: a _____ system that controls all parts of your _____.

The two main parts of the human nervous system:

 _____ nervous system (CNS).

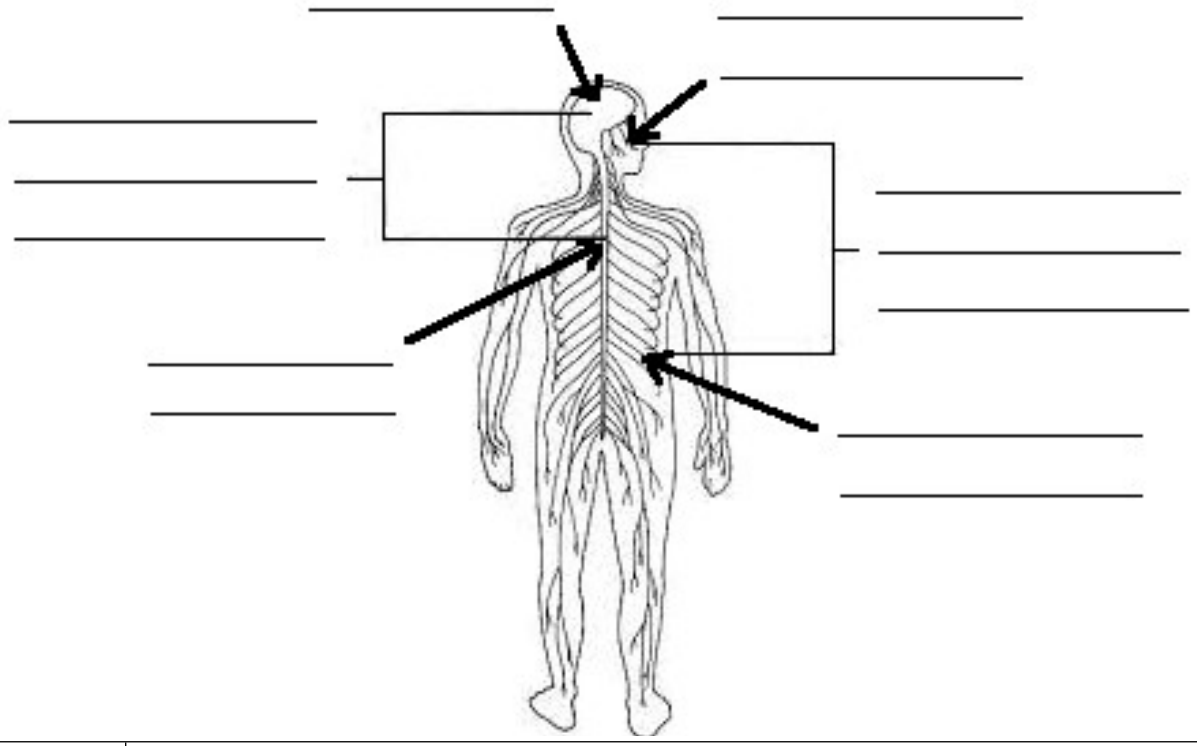
- Made up of the _____ and the spinal cord.

- Receives information from all over the body, _____ the information and sends out messages telling the body how to _____.

 Peripheral _____ system (PNS).

- Made up of the _____ that carry messages to and from the CNS and other parts of the body.

Label the diagram below using figure 7.1.1 on page 225.



Term	Definition/description
Central Nervous System	
Peripheral Nervous System	

Nerve cells

➢The nervous system is made up of trillions of _____ cells (neurons).

➢**Neurones**: nerve _____.

➢ Neurones carry electrical _____ (nerve impulses) from one part of your body to another very quickly. The nerve impulses can only travel in one _____.

➢ **Nerve impulses**: the electrical messages carried by a _____ cell.

A neurone has four main parts.

List them below.

_____

➢ **Cell body**: the part of the neurone that contains the _____.

➢ **Dendrites**: branches from the cell body that receive _____ from other neurons.

➢ **Axon**: a nerve fibre that sends nerve _____ away from the cell body.

There are two common types of neurones.

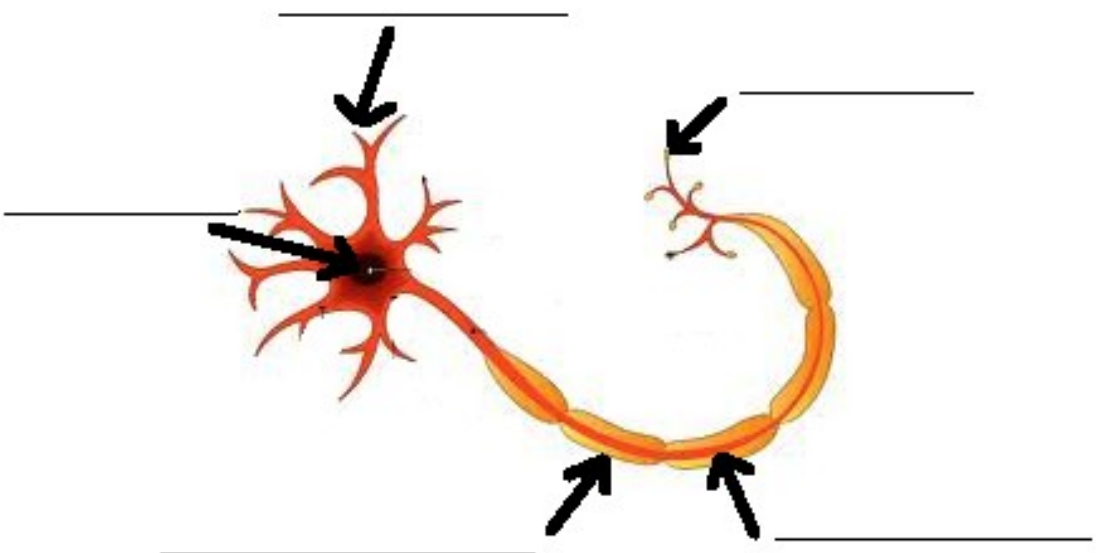
Motor neurones and _____ neurones.

➢ **Motor neurones**: nerve cells that carry messages from the _____to effectors.

➢ **Effectors**: muscles or glands that put messages into _____.

➢ **Myelin sheath**: the insulating layer that covers a _____.

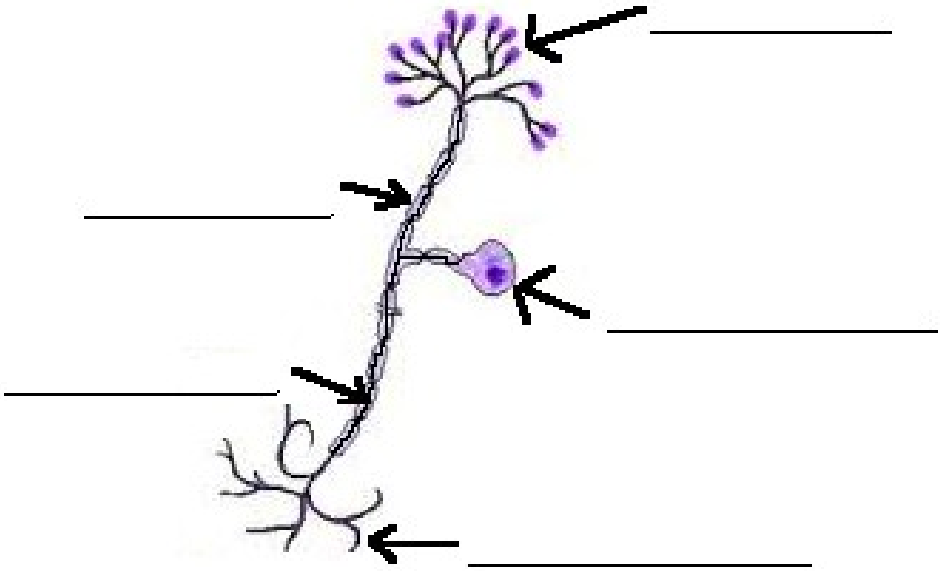
Motor neurone diagram



➢ **Sensory neurones** are:_____

_ _____

Sensory neurone diagram



1. Explain the difference between the cell body in the motor neurone compared to the cell body in the sensory neurone.

_____

_____

_____

2. What is the insulating covering around the axon called?

_____

Why is it needed? _____

_____

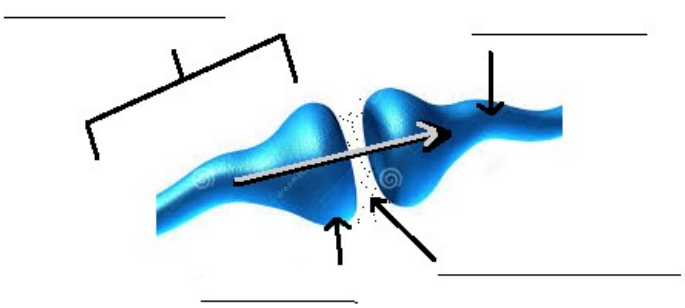
3. What is located inside the cell body?

_____

➢ **Neurotransmitter**: a _____ that sends signals from one neurone to another.

➢ **Synapse**: the space between two _____.

Label the following diagram illustrating a synapse between two neurones.



Year 9 Biology Homework #2

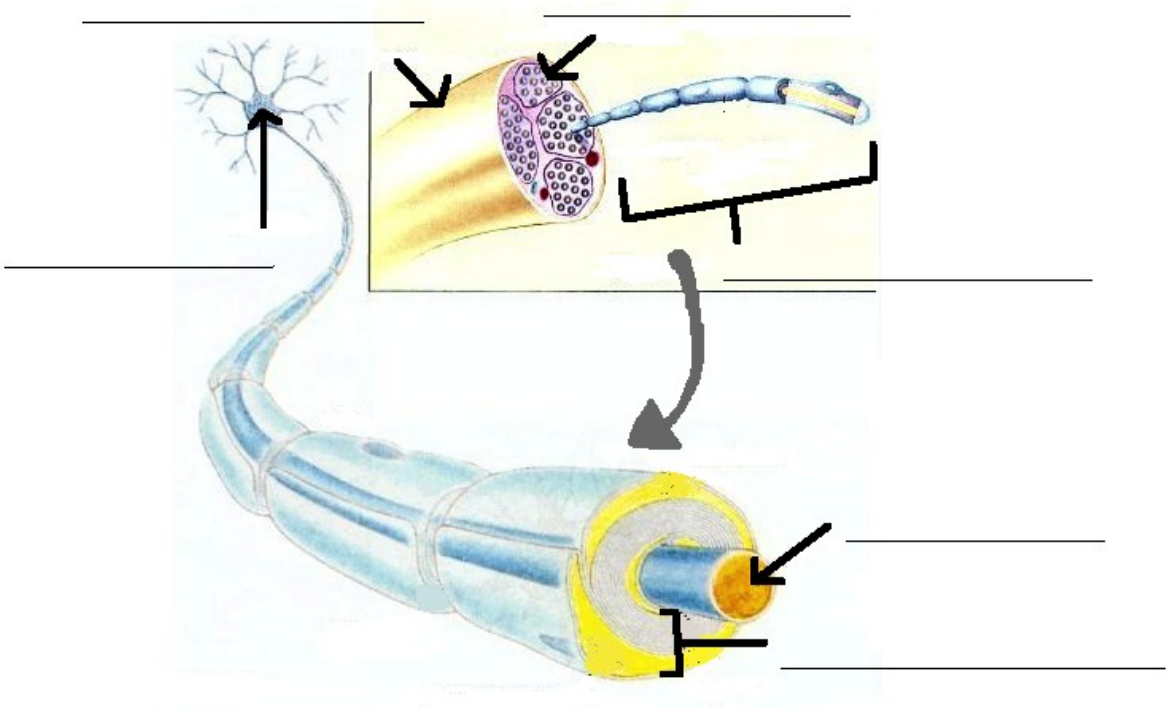
Name:_____

Due_____

Term	Definition/description	Diagram/example
Neurones		
Nerve impulses		
Cell body		
Dendrites		
Axon		
Motor neurone		
Effectors		

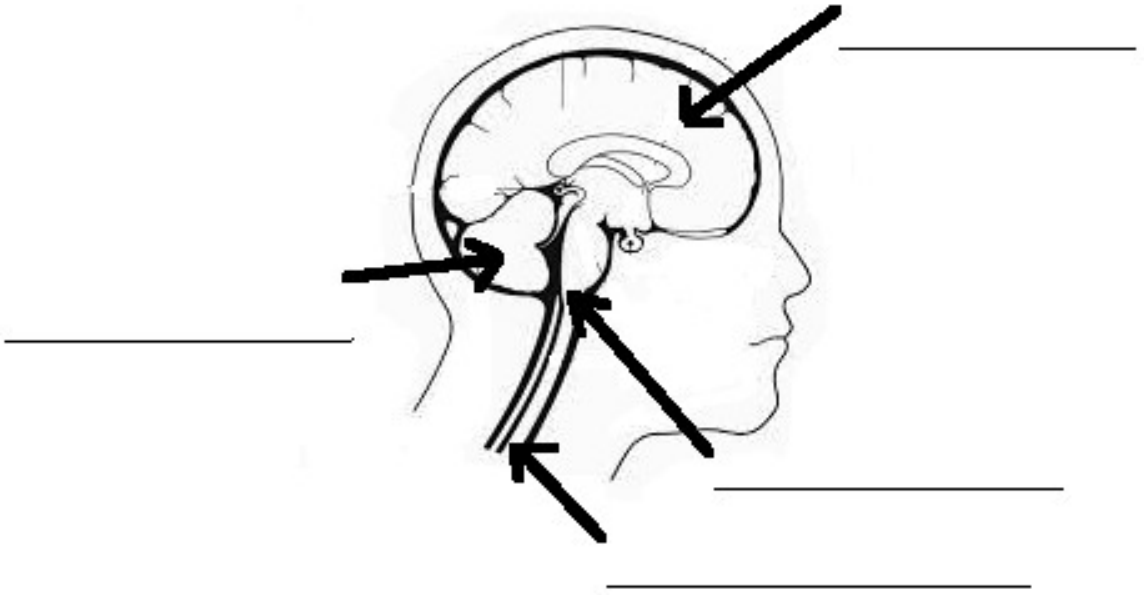
Myelin sheath		
Term	Definition/description	Diagram/example
Sensory neurones		
Neuro-transmitter		
Synapse		

Diagram showing nerve



The brain

➢



Cerebrum: part of the brain that controls conscious thoughts, controls the movement and receives sensory messages from each body part.

The left hemisphere controls the _____ side of the body.

The right hemisphere controls the _____ side of the body.

➢**Cerebellum**: part of the brain responsible for coordination and balance.

➢ **Brain stem**: part of the brain where the spinal cord enters the skull.

➢ **Medulla**: the lower half of the brain _____.

Term	Definition/description
Cerebellum	
Brain stem	
Medulla	

Year 9 Biology Homework #3

Name:_____

Due_____

PERIPHERAL NERVOUS SYSTEM

There are two parts of the peripheral (PNS).

➢The somatic nervous system.

➢The _____ nervous system.

Somatic nervous system

➢ **Somatic nervous system**: part of the nervous system that coordinates the movement of the body and receives information from the sensory _____.

➢ **Receptors**: special cells that detect _____.

➢ **Stimulus**: any factor that stimulates a receptor and causes a _____.

➢ **Optic nerve**: the nerve that carries messages from the _____ to the brain.

1. Explain how the skin on your body protects itself from the outside world.

_____

_____

2. Explain how the receptors in your ear helps you to keep your balance.

_____

_____

_____

3. Explain what chemoreceptors are and where they are located.

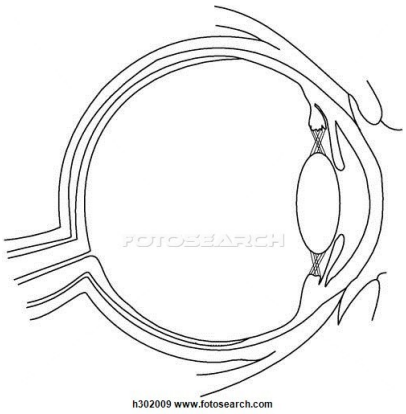
_____

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_____

What are photoreceptors? _____

_



Locate and label the part of the eye where photoreceptors are located

There are two types in the human eye. Name each and describe their function._____

_

Term	Definition/description	Diagram/example
Somatic Nervous System		
Receptors		
Stimulus		

Autonomic nervous system

- **Autonomic nervous system**: the system controlling _____ actions.

- **Sympathetic nervous system**: part of the nervous system that speeds up the functions of the body and makes the body work more _____.

List seven functions that the sympathetic nervous system has. (Page 230 text book).

_____

_____

_____

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➢ **Parasympathetic nervous system**: part of the nervous system that _____ the body down and controls it when it is resting.

List five functions that the parasympathetic nervous system has. (Page 231 text book).

_____

_____

_____

_____

_____

1. Compare the main difference between the sympathetic nervous system and the parasympathetic nervous system.

Term	Definition/description	Diagram/example
Autonomic Nervous System		
Sympathetic Nervous System		
Para-sympathetic Nervous System		

Responding to stimuli

➢A stimulus response model is a simple model of your _____ system.

➢_____ stimulate the sensory nerves and these nerves send a message to the brain.

➢The brain works out the _____ that is required then sends a message along motor nerves to the effectors (these are the muscles or glands that will put the response into effect).

Look at figure 7.1.15 as an example of a stimulus response model.

➢If you touch something _____ or sharp, you automatically pull your hand away.

➢Reflex action protects your body from danger.